

Ex. 3.

Observation Table					
Observation Number	x	y	x*y	x ² *y	x ³
0	0	0	0	0	0
1	1	6	6	6	1
2	2	11	22	44	8
3	3	18	54	162	27
4	4	27	108	432	64
Σ	10	62	190	644	100

1	$y = a + bx + cx^2$
2	$\sum y = na + b\sum x + c\sum x^2$
3	$\sum xy = a\sum x + b\sum x^2 + c\sum x^3$
4	$\sum x^2y = a\sum x^2 + b\sum x^3 + c\sum x^4$

$62 = 4a + 10b + 30c$	eq - 1
$190 = 10a + 30b + 100c$	eq - 2
$644 = 30a + 100b + 354c$	eq - 3

According to Cramer's Rule

Del = $\begin{vmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{vmatrix}$	A = $\begin{vmatrix} 62 & 10 & 30 \\ 190 & 30 & 100 \\ 644 & 100 & 354 \end{vmatrix}$	C = $\begin{vmatrix} 4 & 10 & 62 \\ 10 & 30 & 190 \\ 30 & 100 & 644 \end{vmatrix}$
Del = MDETERM(E33:G35)	B = $\begin{vmatrix} 4 & 62 & 30 \\ 10 & 190 & 100 \\ 30 & 644 & 354 \end{vmatrix}$	: MDETERM(I33:L35) = 80
$\boxed{\text{Del} = 80}$	: MDETERM(J29:L31) = 160	: MDETERM(J25:L27) = 240
x = $\begin{vmatrix} 80 & 190 & 644 \end{vmatrix}$		

a =	A/Del	(M26/D37)	3
b =	B/Del	(M30/d37)	2
c =	C/Del	(M34/D37)	1

>>>> From this values The equation of Secondary Polynomial
Put values of a , b and c in eq1

The Secondary Polynomial : $y = x^2 + 2x + 3$

